

Comparative Analysis of I2P and Tor Network Architectures: Impact on Bandwidth and Latency Performance

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Abstract:

Peer-to-peer networking has gained prominence for its decentralised architecture, providing anonymity and reducing reliance on centralised servers. However, P2P systems face critical performance issues, notably higher latency and suboptimal routing paths. This paper presents a technical evaluation of the network architectures of I2P and Tor, focusing on the effects of architectural decisions on latency and bandwidth. We conducted tests to measure protocol performance using HTTP-GET requests and full website loads, analysing how different architectural choices—such as unidirectional vs. bidirectional tunnels and encryption schemes—impact network efficiency. The findings suggest that while I2P demonstrates marginal advantages over Tor for basic HTTP requests, its performance significantly degrades for more complex website navigation and file downloads.

1. Introduction

The term Peer-To-Peer (P2P) was first popularised in the late 1990s and early 2000s. During this time, file sharing services including Napster highlighted the possibilities of sharing digital content directly between users rather than relying on a central server. The first P2P network was the early internet, with FTP and telnet being predominant applications while each device was a potential source and destination of the application's flows. This flow model can also describe scenarios where all users in a group need equal access to each other for an application. Besides the previously mentioned file-sharing and remote access applications, it can apply to multimedia and tele-services applications (e.g., teleseminars, telelearning, teleconferencing), where any participant can send or receive data to or from any other participant. While these tele-services often function as one-to-many applications, each includes

components that enable one-on-one interactions [1].

2. Network Architecture of I2P and Tor

Anonymity networks like the Invisible Internet Project (I2P) and The Onion Router (Tor) rely on unique routing protocols to provide secure and anonymous communication by passing traffic through multiple peers or “nodes”. However they have different methods and focuses for achieving anonymity.

Tor utilises a process known as onion routing, a decentralised overlay network aimed at anonymizing TCP-based applications such as web browsing, secure shell, and instant messaging [2]. Data is encrypted and sent through a layered network of nodes. Each node is a separate user connected to the Tor network and has different functions depending on how the circuit is relayed. Onion routing is built off the concept of multi-hop routing, a technique of sending data packets through multiple intermediate nodes (or hops) before reaching their final destination. A path is chosen through the network by the client and a circuit is built where nodes connect to one another while only accessing data from their predecessor and successor connections but no others in the circuit [2]. These nodes take the following roles:

Entry Node (Guard Node) - Knows the user's IP address but not the final destination

Relay Node (Middle Node) - Only knows the IP address of the entry and exit node

Exit node - Knows the destination IP address but not the user's

Traffic is sent through 514 byte cells which are decrypted by a symmetric public key called the "onion key". Overall, onion routing ensures that no single node can link the sender to the recipient, thus anonymizing traffic to access the public internet while also providing access to hidden services hosted on Tor known as .onion sites.

In contrast, while Tor was designed for anonymous internet browsing, I2P primarily focuses on building an isolated anonymous network with varied applications. I2P utilises the process of garlic routing, a routing system made up of unidirectional paths, or “tunnels”, to transfer data to relevant peers, or “relays”. Due to I2P's tunnels being unidirectional,

two paths are required: an *outbound tunnel* for sending data to the destination and an *inbound tunnel* for receiving data back from the destination. Garlic routing is considered to be an enhanced version of onion routing due to it bundling multiple packets or “cloves” into a single encrypted packet known as the “garlic bulb”. This bundling obscures the origin of individual packet destinations since each clove within the bulb can be individually encrypted and targeted to a specific recipient. Additionally, this allows multiple communications to be routed together, reducing the number of packages on the network. I2P also hosts hidden services known as “eepsites”. Instead of regular IP addresses, eepsites use a base32 encoded I2P address, often represented as an example website.i2p, to access the service without leaving the network or exposing their IP address. Ultimately, I2P’s design revolves around internal communication solely within its anonymous network, distinguishing it from Tor’s structure, which enables broader access to both anonymous internet browsing and hidden services.

3. Experimental Setup

Anonymising traffic through I2P leads to additional latency and bandwidth loss with each taken hop. Through this paper, we aim to measure average latency and bandwidth when accessing websites through Tor compared to I2P. To accomplish this, a testing environment was established with two machines connected to the I2P network. One machine is configured as a dedicated out proxy, while the other client is set up to utilise both the public outproxy and the dedicated proxy.

The group was tasked with quantifying latency and bandwidth objectively. To accomplish this, a set of perl scripts were used in order to measure HTTP GET Request durations (CORE Latency), Duration of downloading whole webpages with images and external resources and Speeds when downloading certain amounts of data (Bandwidth).

These perl scripts were previously written and used by Fabian et al [6] who analysed latency and bandwidth within the Tor network. The scripts were used to emulate user requests for browsing websites and recorded the measured data. A list of the 500 most frequently visited websites were used with each

target being accessed alternately through direct connection and via I2P.

Environment Configuration:

Two computers that were part of the I2P network were used for the testing setup; one was set up as a dedicated out proxy, while the other was set up as a client that used both dedicated and public out proxies. This configuration allowed benchmarking against direct access and previously released Tor data, enabling controlled network performance testing under I2P’s normal operating settings.

Measurement Tools and Techniques:

The researchers employed a set of Perl scripts created by Fabian et al. and confirmed in previous research for assessing latency and bandwidth in Tor networks [5] in order to collect objective data. These scripts were designed to measure data transfer speeds across different file sizes, download entire web pages with images and external resources, and send HTTP GET requests automatically. This method yielded accurate bandwidth (as download speed) and latency (as HTTP GET request durations) data.

Testing Protocols

A sample of 500 of the most popular websites were tested, and each one was accessed via Tor, I2P, and a direct connection. The I2P client was built up using a 2-hop tunnel and standard parameters for normal usage scenarios. To obtain a representative sampling of network conditions, testing was conducted using both dedicated and public out proxies alternately. The consistency of performance measurement was then ensured by comparing Tor results to published benchmarks [6].

This experimental technique sheds light on the architectural implications on user experience in anonymous web browsing and data transfer situations by offering a methodical way to evaluate the effects of I2P’s unidirectional tunnels and Tor’s bidirectional circuits on latency and bandwidth.

4. Measurement Techniques

Our methodology for analysing I2P and Tor's performance concentrated on measuring three essential network performance metrics: bandwidth, average latency, and core latency. Automated Perl scripts that were previously verified in Fabian et al.'s work on Tor were used for the measurements. These scripts used controlled HTTP-GET requests and website loading simulations to offer accurate measures of network performance. [6]

The following key metrics were captured:

Core Latency:

The baseline response time recorded for straightforward HTTP-GET queries is known as core latency. This measure shows how long it takes for a single HTTP request to travel across both networks and be received. Response timings were noted after each test sent out an HTTP-GET request for a target URL. By comparing instantaneous responses between I2P and Tor, we were able to isolate fundamental delay concerns without requiring the loading of extra resources. When comparing the intrinsic delays brought about by the routing architecture of each network—specifically, the unidirectional tunnels of I2P with the bidirectional circuits of Tor—this latency component is crucial.

Average Latency:

Average latency measurements included the time needed to load complete webpages, including all auxiliary resources like graphics and scripts, to evaluate real-world browsing experiences. This test revealed the compounding delays users may experience by simulating typical browsing activity. It was anticipated that the complexity of page loading would increase latency because I2P relies on numerous hops within distinct incoming and outgoing tunnels. The time it took for 500 of the most popular websites to load was noted for each test, with direct access, I2P, and Tor configurations being alternated.

Bandwidth:

Our bandwidth measurements focused on the speed at which files of different sizes were retrieved to

estimate download efficiency. A range of bandwidth performance across tiny to large files was possible by downloading files in increments of 50 KB to 1 MB. The ability of each network to manage data-intensive jobs was demonstrated by this incremental method, especially when the encryption cost imposed by Tor's onion routing and I2P's garlic routing was taken into account.

By using alternate proxies and constant test cycles, this methodology ensured data integrity while offering an organized way to analyze latency and bandwidth objectively. To find differences and patterns among the networks, the results were statistically examined using mean, median, and quartile computations.

The testing used a combination of open-source Perl scripts, simulating real-world browsing behaviour to maintain uniformity.

5. Results and Analysis

This section will retain the paper's key data findings, but with a more technical and analytical tone:

The I2P and Tor networks are compared in this section with an emphasis on bandwidth throughput, average latency, and core latency. The analyses point to significant variations between the two networks, which are explained by different design decisions such as unidirectional versus bidirectional tunnelling and encryption techniques. In order to guarantee accurate cross-comparisons, these studies make use of earlier techniques used by Fabian et al. for assessing Tor network latency and bandwidth, which have been modified here to evaluate I2P under comparable circumstances [6].

Core Latency

The first test conducted was a comparison of the *core latency* between I2P and tor-loading websites.

The main metric for evaluating response efficiency in anonymity networks is core latency, which measures the amount of time needed to finish a simple HTTP-GET request. The results show a significant

performance difference, with Tor recording a much lower average core latency of 3.31 seconds compared to I2P's average of 10.07 seconds. The inherent overhead introduced by I2P's unidirectional tunnel architecture, which requires distinct inbound and outbound channels for every request, is reflected in this discrepancy. This architecture decision is directly responsible for the higher latency in I2P as unidirectional channels necessitate individual pathways, which lengthens reaction times and increases the likelihood of congestion, especially in situations of contrast. Tor's bidirectional tunnelling minimises the requirement for repeated channel construction and teardown by handling both inbound and outgoing communication within a single tunnel. Tor's fundamental latency is greatly decreased by its effective routing structure, which makes it more appropriate for situations involving frequent, low-overhead requests. These findings are consistent with earlier research that attributes Tor's improved HTTP-GET request processing performance to its simplified bidirectional design [6].

I2P showed an average core latency of 10.07 seconds compared to Tor's previously recorded average of 3.31 seconds. The increased delay can be attributed to I2P's unidirectional tunnel design, which introduces a greater overhead per request. In contrast, Tor's bidirectional tunnels demonstrate more efficiency for simple HTTP requests.

Average Latency

An even greater disparity between the networks may be seen in average latency, which measures how long it takes for online pages with multimedia components to fully load. With latency values peaking at over 16 times that of direct access, I2P's latency performance is noticeably worse. In particular, the 25-quantile was 14.39 seconds and the 75-quantile was 226.85 seconds, whereas the median latency for I2P was over 103.19 seconds. Tor, on the other hand, had a more consistent latency distribution, with 25 and

75-quantile values at 9.60 and 30.26 seconds, respectively. This discrepancy demonstrates how well Tor handles requests involving large amounts of data, in contrast to I2P's design, which suffers from delay compounding since it depends on several unidirectional tunnels. I2P is unable to effectively handle multi-resource requests due to the lengthy setup time and overhead involved in creating separate tunnels for incoming and outgoing traffic. Additionally, routing choices are constrained by the scarcity of I2P out proxies, which may cause queue delays as users compete for scarce proxy resources. I2P's latency performance is further deteriorated by this outproxy bottleneck, which affects usability, especially for applications that call for quick access to complicated, resource-intensive information [6].

Bandwidth

Tor's higher bandwidth over I2P is demonstrated by bandwidth tests, which were carried out using incremental file downloads ranging from 50 KB to 1 MB. With an average download speed of 31.28 kB/s and a maximum recorded speed of just 277.82 kB/s, I2P trailed well behind Tor in these tests, which saw Tor reach an average download rate of 937.90 kB/s. Faster, more reliable data transfer is made possible by Tor's bidirectional tunnels, which eliminate the extra setup and routing costs associated with I2P's unidirectional tunnels. I2P's bandwidth limitations are made worse by its garlic encryption technique, which raises the network's computational load even if it is useful for anonymizing several data packets in a single transfer. The lengthier routing pathways of unidirectional tunnels and this encryption overhead restrict I2P's capacity to effectively handle bigger files. Furthermore, I2P's lack of out proxies further restricts throughput during periods of high usage, making it difficult to efficiently handle high-bandwidth operations like streaming or downloading big files. According to earlier research, I2P's architectural choices are the cause of its bandwidth constraints, especially when it comes to intricate, data-intensive applications [6].

Tor's design clearly outperforms I2P in terms of bandwidth, average latency, and core latency. Tor can

sustain faster throughput and lower latency across a variety of workloads thanks to its bidirectional tunnels, effective onion routing, and wider out proxy architecture. In contrast, jobs needing quick access to external data sources or fast download rates suffer from higher latency and lower bandwidth due to I2P's architecture, which places a higher priority on isolated communication within its network. While I2P is still more successful within its intended scope of isolated, anonymous networking, our investigation highlights the applicability of Tor for applications that need low latency and large bandwidth, notably in anonymous web surfing and data-heavy interactions [6].

6. Discussion: Architectural Impact on Network Protocol Performance

This section examines the architectural decisions made by Tor and I2P, emphasizing the effects these designs have on bandwidth and latency in real-world situations. We determine how each network's configuration affects performance by looking at tunnel structure, routing protocols, and encryption overhead.

6.1 Tunnel Routing and Length

The tunnel architectures and routing strategies used by I2P and Tor differ significantly from one another. I2P uses unidirectional tunnels, meaning that in order to finish a communication loop, separate inbound and outgoing channels are needed. By adding more hops, this design raises the number of possible failure spots and adds to cumulative latency. I2P's performance degrades when each relay in the unidirectional tunnel adds routing overhead, particularly when there are fewer outproxy alternatives and higher loads. In contrast, Tor uses bidirectional circuits, in which a single channel manages both outgoing and incoming data. In their latency investigation of Tor's bidirectional routing paradigm, Fabian et al. showed that this reduces the overhead from tunnel formation, lowering latency for individual HTTP requests [6].

The performance disparities between the two networks are further highlighted by the effect of outproxy availability. Congestion brought on by I2P's fewer out proxies can seriously impair performance during periods of high usage. Previous research comparing the scalability of I2P and Tor networks has found that this constraint also affects bandwidth efficiency, particularly for data-intensive applications [6][1].

6.2 Encryption Overhead

Notable performance differences between I2P and Tor can also be attributed to their encryption strategies. Several data packets, or "cloves," are combined into a single message (the "garlic bulb") for transmission when using I2P's garlic routing technique. The intricacy of handling numerous encrypted cloves raises processing costs at each relay node, despite the fact that garlic routing is theoretically effective for combining data and adding anonymity layers. Delay is introduced by this additional complexity, especially when there are large packet volumes or lengthy transmission channels inside the network [1].

On the other hand, Tor's onion routing was designed to have as little delay as possible, especially for regular HTTP traffic. As data moves between nodes, onion routing applies multilayer encryption in a sequential manner, enabling each node to decrypt only one layer before sending the data on to the next node. Throughput is increased and total delay is decreased with this simplified encryption model, which uses fewer processing resources at each hop. Dingledine et al.'s empirical research, which attributes Tor's effectiveness to the ease of onion routing, further supports its performance advantage in common browsing scenarios [2].

6.3 Comparative Analysis

The trade-offs included in each network's architecture are highlighted by the interaction between encryption complexity and tunnel design. I2P prioritizes layered security at the expense of increased latency and capacity limitations because it was initially created for safe internal communications within its network. For tasks requiring quick data transfer, such as web

browsing or streaming, where Tor routinely performs better than I2P, this trade-off reduces I2P's effectiveness. This trend is supported by research by Ehlert et al., which demonstrates that Tor performs best in situations requiring effective external network access, like anonymous web browsing, while I2P is best suited for isolated, internal communication tasks [6].

In conclusion, I2P's unidirectional tunnels and garlic routing improve anonymity, but they also increase latency and limit bandwidth. On the other hand, Tor's performance-optimized onion routing and bidirectional tunnels offer a good choice for applications that are sensitive to latency. By bridging the gap between these architectural paradigms, future improvements like hybrid routing or improved outproxy support may enhance I2P's performance for larger use cases [5][4].

7. Conclusion: Design Trade-offs in Anonymity Networks

Significant architectural trade-offs between I2P and Tor are revealed by this study, and each is appropriate for a particular use case involving anonymity. Strong internal security is provided by I2P's unidirectional tunnels and garlic routing, which makes it useful for anonymous peer-to-peer interactions and isolated network communications. Nevertheless, its unidirectional tunnel design, which necessitates distinct pathways for data entering and leaving, adds latency and bandwidth constraints, especially affecting its capacity to manage high-traffic situations like web browsing. This limits I2P's effectiveness for applications that need fast response times and low-latency data transfers, as previous network performance studies have pointed out [1].

Tor, in contrast, optimizes its architecture for greater accessibility and reduced latency, particularly when accessing external web resources, by utilizing bidirectional circuits and onion routing. Tor reduces the number of hops per transaction by managing both inbound and outbound traffic within a single circuit, which lowers latency and makes it easier to scale in

situations with high demand. Tor is especially well-suited for situations that demand both strong anonymity and excellent network responsiveness, such as anonymous web browsing, because of its vast network of relays, which further promotes reliable performance [2].

In conclusion, Tor works well for performance-demanding apps that need external access, while I2P's architecture is more suited for applications that prioritize private, secure communications over an isolated network. To overcome these trade-offs, future studies could look into hybrid strategies or adaptive routing, which could improve I2P's suitability for latency-sensitive situations while maintaining its security benefits. [6] [4].

8. Future Work

Key areas for improving I2P and Tor performance might be the subject of future study, with an emphasis on architectural optimisations to increase bandwidth and latency. Future research may look into scaling methods to expand the number and geographic dispersion of outproxies in light of I2P's present outproxy availability limits. This might potentially improve latency and reduce congestion during times of high demand [1]. Investigating adaptive tunnelling techniques in I2P may also aid in striking a balance between security and latency by dynamically modifying tunnel lengths in response to network circumstances, improving efficiency without sacrificing anonymity. Additionally, this might help I2P become more competitive with Tor in managing high-throughput activities by reducing its performance deterioration for intricate browsing and huge file transfers [3], [4].

It could also be advantageous to optimise encryption techniques. Evaluating simplified encryption techniques or hybrid encryption models that use onion and garlic routing capabilities, for example, may lower the overhead seen in both networks while preserving strong security. I2P may become more feasible for bandwidth-intensive operations like

multimedia streaming and large-scale downloads if this strategy lowers processing needs [5].

Future developments on Tor may concentrate on making it more user-friendly by enhancing its circuit-building algorithms to increase speed and lower latency even more, particularly in areas with few network access points [2], [6]. Novel ways to strike a compromise between security and performance may be provided by hybrid techniques that take advantage of the advantages of both I2P and Tor architectures. For example, selectively implementing unidirectional tunnels within Tor for particular applications. Furthermore, using machine learning models for routing optimisation and traffic prediction may improve both networks' ability to adjust to changing user demands by cutting down on delays and improving data flow efficiency [3].

Similar to Tor's DNS, including a decentralised name service into I2P might speed up address resolution and facilitate network navigation. This functionality may improve I2P's usability for web-browsing apps by bridging the gap between private and public browsing settings and expanding I2P's use cases beyond its present scope [4], [5].

<http://www.i2p2.de/en/docs/how/tech-intro> [Accessed 11 October 2024].

[6] Fabian, B., Goertz, F., Kunz, S., Müller, S. and Nitzsche, M., 2010. *Privately Waiting – A Usability Analysis of the Tor Anonymity Network*. Proceedings of the Sixteenth Americas Conference on Information Systems, Lima, Peru, August 12-15, 2010.

Bibliography:

- [1] Mack, J. D. (2003). *Network analysis, architecture, and design* (2nd ed.). Morgan Kaufmann.
- [2] Dingledine, R., Mathewson, N., & Syverson, P. (2004). *Tor: The Second-Generation Onion Router*. The Free Haven Project and Naval Research Lab.
- [3] Tor Project. (n.d.). *Key management*. Tor Project Support. Retrieved [date you accessed the page], from <https://support.torproject.org/about/key-management/>
- [4] Astolfi, F., Kroese, J., & van Oorschot, J., 2015. *I2P - The Invisible Internet Project*. Leiden University.
- [5] The Invisible Internet Project, n.d. *Technical Introduction*. Available at: